Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that: The DNS server or port is unreachable, because the network returned an ICMP error message saying "udp port 53 unreachable This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: udp port 53 unreachable The port noted in the error message is used for: DNS (Domain Name server) to find the IP address of this website yummyreciepesforme The most likely issue is: the server or a firewall is rejecting the connection (closing port 53 or blocking the traffic), and because the traffic volume is very low, it is definitely not a DoS attack

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time incident occurred: First attempt: 13:24:32 (with the ICMP unreachable error returned at 13:24:36) Second attempt: 13:26:32 (with the ICMP unreachable error returned at 13:27:15) Third attempt: 13:28:32 (with the ICMP unreachable error returned at 13:28:50) Explain how the IT team became aware of the incident: The IT team or monitoring systems noticed repeated ICMP error messages ("udp port 53 unreachable") coming back from the network when trying to resolve domain names. Explain the actions taken by the IT department to investigate the incident: The IT department captured and analyzed network traffic logs (tcpdump) to track communication attempts between the client and the DNS server over several minutes. Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): Affected Port: UDP Port 53 (used for DNS services). DNS Server IP: 203.0.113.2.

Client IP: 192.51.100.15.
Target Domain: yummyrecipesforme.com.
Error Message: ICMP udp port 53 unreachable returned in response to the DNS queries.

Note a likely cause of the incident: The DNS server is turned off, offline, or a firewall is actively blocking or rejecting incoming traffic on UDP port 53.